ANNEXURE-III

TECHNICAL SPECIFICATION FOR

HARDWARE FITTINGS AND ACCESSORIES OF

HIGH PERFORMANCE CONDUCTORS

(AAAC PANTHER CONDUCTORS)

TECHNICAL SPECIFICATION FOR HARDWARE FITTINGS AND ACCESSORIES SUITABLE FOR AAAC PANTHER (HP conductors)

1. Technical Description of Hardware Fittings

1.1 General

This section details technical particulars of fittings viz. suspension clamps and compression type dead end clamps for the AAAC Panther Conductor to be supplied by the bidder. Each fitting shall be supplied complete in all respects.

1.2 The fittings shall be suitable for attachment to suspension and tension insulator strings along with hardware fittings and shall include 2.5 % extra fasteners and Aluminium filler plugs. The supplier shall be responsible for satisfactory performance of complete conductor system along with fittings offered by them for continuous operation at the designed maximum temperature specified by them for the conductor.

1.3 Corona and RI Performance [for 400kV]

Sharp edges and scratches on all the hardware fittings shall be avoided. All surfaces must be clean, smooth, without cuts and abrasions or projections. The Supplier shall be responsible for satisfactory corona and radio interference performance of the materials offered by him.

1.4 Maintenance

1.4.1 The hardware fittings offered shall be suitable for employment of hot line maintenance technique so that usual hot line operations can be carried out with ease, speed and safety. The technique adopted for hot line maintenance shall be generally bare hand method & hot stick method.

1.5 Split Pins

1.5.1 Split pins shall be used with bolts & nuts.

1.6 Suspension Assembly

1.6.1 The suspension assembly shall be suitable for the AAAC Panther Conductor, the bidder intends to supply. The technical details of the conductor shall be as proposed by the bidder.

1.6.2 The suspension assembly shall include either free center type suspension clamp along with standard preformed armour rods or armour grip suspension clamp.

1.6.3 The suspension clamp along with standard preformed armour rods set shall be designed to have maximum mobility in any direction and minimum moment of inertia so as to have minimum stress on the conductor in the case of oscillation of the same.

1.6.4 The suspension clamp suitable for various type of Conductor along with standard preformed armour rods/armour grip suspension clamp set shall have a slip strength in conformity with relevant Indian/ International standards.

1.6.5 The suspension clamp shall be designed for continuous operation at the temperature specified by the bidder for conductor.

1.6.6 The suspension assembly shall be designed, manufactured and finished to give it a suitable shape, so as to avoid any possibility of hammering between suspension assembly and conductor due to vibration. The suspension assembly shall be smooth without any cuts, grooves, abrasions, projections, ridges or excrescence which might damage the conductor.

1.6.7 The suspension assembly/clamp shall be designed so that it shall minimize the static & dynamic stress developed in the conductor under various loading conditions as well as during wind induced conductor vibrations.

1.6.8 It shall also withstand power arcs & have required level of Corona/RIV performance.

1.6.9 Free Centre Type Suspension Clamp

For the Free Centre Suspension Clamp seat shall be smoothly rounded and curved into a bell mouth at the ends. The lip edges shall have rounded bead.

There shall be at least two U-bolts for tightening of clamp body and keeper pieces together.

1.6.10 Standard Preformed Armour Rod Set

1.6.10.1 The Preformed Armour Rods Set shall be used to minimize the stress developed in the sub-conductor due to different static and dynamic loads because of vibration due to wind, slipping of conductor from the suspension clamp as a result of unbalanced conductor tension in adjacent spans and broken wire condition. It shall also withstand power arcs, chafing and abrasion from suspension clamp and localized heating effect due to magnetic power losses from suspension clamps as well as resistance losses of the conductor.

1.6.10.2 The preformed armour rods set shall have right hand lay and the inside diameter of the helics shall be less than the outside diameter of the conductor to have gentle but permanent grip on the conductor. The surface of the armour rod when fitted on the conductor shall be smooth and free from projections, cuts and abrasions etc.

1.6.10.3 The pitch length of the rods shall be determined by the Bidder but shall be less than that of the outer layer of conductor and the same shall be accurately controlled to maintain uniformity and consistently reproducible characteristic wholly independent of the skill of linemen.

1.6.10.4 The length and diameter of each rod shall be furnished by the bidder in the GTP. The tolerance in length of the rods between the longest and shortest rod in complete set should be within the limits specified in relevant Indian/International Standards. The ends of armour rod shall be parrot billed.

1.6.10.5 The length and diameter of each rod shall be specified in the GTP. The tolerance in length of the rods in complete set should be within 13 mm between the longest and shortest rod. The ends of armour rod shall be parrot billed.

1.6.10.6 The number of armour rods in each set shall be supplier's design to suit AAAC Panther conductor offered Standards. Each rod shall be marked in the middle with paint for easy application on the line.

1.6.10.7 The armour rod shall not lose their resilience even after five applications.

1.6.10.8 The conductivity of each rod of the set shall not be less than 40% of the conductivity of the International Annealed Copper Standard (IACS).

1.6.11 Armour Grip Suspension Clamp

1.6.11.1 The armour grip suspension clamp shall comprise of retaining strap, support housing, elastomer inserts with aluminium reinforcements and AGS preformed rod set.

1.6.11.2 Elastomer insert shall be resistant to the effects of temperature up to designed maximum conductor temperature guaranteed by the bidder corresponding to peak current, Ozone, ultraviolet radiations and other atmospheric contaminants likely to be encountered in service. The physical properties of the elastomer shall be of approved standard. It shall be electrically shielded by a cage of AGS performed rod set. The elastomer insert shall be so designed that the curvature of the AGS rod shall follow the contour of the neoprene insert.

1.6.11.3 The supplier shall submit relevant type/performance test certificates as per applicable standard/product specifications for elastomer to confirm suitability of the offered elastomer for the specified application.

1.6.11.4 The AGS preformed rod set shall be as detailed in clause 1.6.10.4 to 1.6.10.7 in general except for the following.

1.6.11.4 The length of the AGS preformed rods shall be such that it shall ensure sufficient slipping strength as detailed under clause 1.6.4 and shall not introduce unfavourable stress on the conductor under all operating conditions. The length of the AGS preformed rods shall be indicated in the GTP.

1.7 Envelope Type Suspension Clamp

1.7.1 The seat of the envelope type suspension clamp shall be smoothly rounded & suitably curved at the ends. The lip edges shall have rounded bead. There shall be at least two U-bolts for tightening of clamp body and keeper pieces together. Hexagonal bolts and nuts with split-pins shall be used for attachment of the clamp.

1.8 Dead end Assembly

1.8.1 The dead end assembly shall be suitable for the AAAC Panther Conductor.

1.8.2 The dead end assembly shall be of compression type with provision for compressing jumper terminal at one end. The angle of jumper terminal to be mounted (including angle of pad) should be 30° with respect to the vertical line. The area of bearing surface on all the connections shall be sufficient to ensure positive electrical and mechanical contact and avoid local heating due to I2R losses. The resistance of the clamp when compressed on Conductor shall not be more than 75% of the resistance of equivalent length of Conductor.

1.8.3 Die compression areas shall be clearly marked on each dead-end assembly designed for continuous die compressions and shall bear the words 'COMPRESS FIRST' suitably inscribed near the point on each assembly where the compression begins. If the dead end assembly is designed for intermittent die compressions it shall bear identification marks 'COMPRESSION ZONE' AND 'NON-COMPRESSION ZONE' distinctly with arrow marks showing the direction of compressions and knurling marks showing the end of the zones. Tapered aluminium filler plugs shall also be provided at the line of demarcation between compression & non-compression zone. The letters, number and other markings on the finished clamp shall be distinct and legible. The dimensions of dead end assembly before & after compression along with tolerances shall be guaranteed in the relevant schedules of the bid and shall be decided by the manufacturer so as to suit the conductor size & conform to electrical & mechanical requirement stipulated in the specification. These shall be guaranteed in the relevant schedules of bid.

1.8.4 The assembly shall not permit slipping of, damage to, or failure of the complete conductor or any part thereof at a load less than 95% of the ultimate tensile strength of the conductor.

1.8.5 Jumper bolting arrangement between jumper terminal/cone and terminal pad/plate of dead end assembly of tension hardware fittings shall be designed to suit the specification requirement of ---A current and shall conform to the relevant Indian/International standards

1.8.6 Deleted

1.9 Fasteners: Bolts, Nuts and Washers

1.9.1 All bolts and nuts shall conform to IS 6639. All bolts and nuts shall be galvanized as per IS 1367 (Part-13)/IS 2629. All bolts and nuts shall have hexagonal heads, the heads being forged out of solid truly concentric, and square with the shank, which must be perfectly straight.

1.9.2 Bolts up to M16 and having length up to 10 times the diameter of the bolt should be manufactured by cold forging and thread rolling process to obtain good and reliable mechanical properties and effective dimensional control. The shear strength of bolt for 5.6 grade should be 310 MPa minimum as per IS 12427. Bolts should be provided with washer face in accordance with IS 1363 (Part-1) to ensure proper bearing.

1.9.3 Nuts should be double chamfered as per the requirement of IS 1363 Part-III 1984. It should be ensured by the manufacturer that nuts should not be over tapped beyond 0.4 mm oversize on effective diameter for size up to M16.

1.9.4 Fully threaded bolts shall not be used. The length of the bolt shall be such that the threaded portion shall not extend into the place of contact of the component parts.

1.9.5 All bolts shall be threaded to take the full depth of the nuts and threaded enough to permit the firm gripping of the component parts but no further. It shall be ensured that the threaded portion of the bolt protrudes not less than 3 mm and not more than 8 mm when fully tightened. All nuts shall fit and tight to the point where shank of the bolt connects to the head.

1.9.6 Flat washers and spring washers shall be provided wherever necessary and shall be of positive lock type. Spring washers shall be electro-galvanized. The thickness of washers shall conform to IS:2016.

1.9.7 The supplier shall furnish bolt schedules giving thickness of components connected, the nut and the washer and the length of shank and the threaded portion of bolts and size of holes and any other special details of this nature.

1.9.8 To obviate bending stress in bolt, it shall not connect aggregate thickness more than three time its diameter.

1.9.9 Bolts at the joints shall be so staggered that nuts may be tightened with spanners without fouling.

1.9.10 To ensure effective in-process Quality control it is essential that the manufacturer should have all the testing facilities for tests like weight of zinc coating, shear strength, other testing facilities etc., in-house. The manufacturer should also have proper Quality Assurance system which should be in line with the requirement of this specification and IS-.14000 services Quality System standard.

1.9.11 Fasteners of grade higher than 8.8 are not to be used and minimum grade for bolt shall be 5.6.

1.10 Materials

The materials of the various components shall be as specified hereunder. The Bidder shall indicate the material proposed to be used for each and every component of hardware fittings stating clearly the class, grade or alloy designation of the material, manufacturing process & heat treatment details and the reference standards.

1.10.1 The details of materials for different component are listed as in Table No-1.

1.11 Workmanship

1.11.1 All the equipment shall be of the latest design and conform to the best modern practices adopted in the Extra High Voltage field. The Bidder shall offer only such equipment as guaranteed by him to be satisfactory and suitable for ---kV transmission lines and will give continued good performance. For employer's review of the offered design of clamps/ fittings, the supplier shall submit document/design details of similar type of clamps/ fittings used in past for similar type of HTLS conductor application

1.11.2 High current, heat rise test shall be conducted by the supplier to determine the maximum temperature achieved in different components of fittings under simulated service condition corresponding to continuous operation of conductor at designed maximum temperature. The material of the components should be suitable for continued good performance corresponding to these maximum temperatures. The supplier shall submit relevant type/performance test certificates as per applicable standards/product specifications to confirm suitability of the offered material.

1.11.3 The design, manufacturing process and quality control of all the materials shall be such as to give the specified mechanical rating, highest mobility, elimination of sharp edges and corners to limit corona and radio-interference, best resistance to corrosion and a good finish.

1.11.4 All ferrous parts including fasteners shall be hot dip galvanized, after all machining has been completed. Nuts may, however, be tapped (threaded) after galvanizing and the threads oiled. Spring washers shall be electro galvanized. The bolt threads shall be undercut to take care of the increase in diameter due to galvanizing. Galvanizing shall be done in accordance with IS2629/ IS1367 (Part-13) and shall satisfy the tests mentioned in IS2633. Fasteners shall withstand four dips while spring washers shall withstand three dips of one minute duration in the standard Preece test. Other galvanized materials shall have a minimum average coating of zinc equivalent to 610 gm/sq.m. shall be guaranteed to withstand at least six successive dips each lasting one (1) minute under the standard preece test for galvanizing.

1.11.5 Before ball fittings are galvanized, all die flashing on the shank and on the bearing surface of the ball shall be carefully removed without reducing the dimensions below the design requirements.

1.11.6 The zinc coating shall be perfectly adherent, of uniform thickness, smooth, reasonably bright, continuous and free from imperfections such as flux, ash rust, stains, bulky white deposits and blisters. The zinc used for galvanizing shall be grade Zn 99.95 as per IS:209.

1.11.7 Pin balls shall be checked with the applicable, “GO” gauges in at least two directions. one of which shall be across the line of die flashing, and the other 90^0 to this line. "NO GO" gauges shall not pass in any direction.

1.11.8 Socket ends, before galvanizing, shall be of uniform contour. The bearing surface of socket ends shall be uniform about the entire circumference without depressions of high spots. The internal contours of socket ends shall be concentric with the axis of the fittings as per IS:2486/IEC:120.

The axis of the bearing surfaces of socket ends shall be coaxial with the axis of the fittings. There shall be no noticeable tilting of the bearing surfaces with the axis of the fittings.

1.11.9 In case of casting, the same shall be free from all internal defects like shrinkage, inclusion, blow holes, cracks etc. Pressure die casting shall not be used for casting of components with thickness more than 5 mm.

1.11.10 All current carrying parts shall be so designed and manufactured that contact resistance is reduced to minimum.

1.11.11 No equipment shall have sharp ends or edges, abrasions or projections and cause any damage to the conductor in any way during erection or during continuous operation which would produce high electrical and mechanical stresses in normal working. The design of adjacent metal parts and mating surfaces shall be such as to prevent corrosion of the contact surface and to maintain good electrical contact under service conditions.

1.11.12 All the holes shall be cylindrical, clean cut and perpendicular to the plane of the material. The periphery of the holes shall be free from burrs.

1.11.13 All fasteners shall have suitable corona free locking arrangement to guard against vibration loosening.

1.11.14 Welding of aluminium shall be by inert gas shielded tungsten arc or inert gas shielded metal arc process. Welds shall be clean, sound, smooth, uniform without overlaps, properly fused and completely sealed. There shall be no cracks, voids incomplete penetration, incomplete fusion, under-cutting or inclusions.

Porosity shall be minimized so that mechanical properties of the aluminium alloys are not affected. All welds shall be properly finished as per good engineering practices.

1.12 Bid Drawings

1.12.1 The Bidder shall furnish full description and illustrations of materials offered.

1.12.2 Fully dimensioned drawings of the hardware and their component parts shall be furnished --- copies along with the bid. Weight, material and fabrication details of all the components should be included in the drawings.

1.12.3 All drawings shall be identified by a drawing number and contract number. All drawings shall be neatly arranged. All drafting & lettering shall be legible. The minimum size of lettering shall be 3 mm. All dimensions & dimensional tolerances shall be mentioned in mm.

The drawings shall include:

(i) Dimensions and dimensional tolerance.

(ii) Material, fabrication details including any weld details & any specified finishes & coatings. Regarding material designation & reference of standards are to be indicated.

(iii) Catalogue No.

(iv) Marking

(v) Weight of assembly

(vi) Installation instructions

(vii) Design installation torque for the bolt or cap screw.

(viii) Withstand torque that may be applied to the bolt or cap screw without failure of component parts.

(ix) Installation instructions

(x) The compression die number with recommended compression pressure.

(xi) Placement charts for spacer/spacer damper and damper

(xii) All other relevant technical details

1.12.4 After placement of award, the Supplier shall submit fully dimensioned drawing including all the components in copies to the Purchaser for approval. After getting approval from the Purchaser and successful completion of all the type tests, the Supplier shall submit two more copies of the same drawings to the Purchaser for further distribution and field use at Purchaser's end.

TABLE-1

(Details of Materials)

Sl. No.	Name of item	Material treatment	Process of Standard	Reference	Remark s
1.	Security Clips	Stainless Steel/ Phosphor Bronze	-	AISI 302 or 304-L/ IS-1385	
2.	For Free Centre /Envelope type clamps				
(a)	Clamp Body, Keeper Piece	High Strength Al. Alloy 4600/ LM-6 or 6061/65032	Casted or forged & Heat treated	IS:617or ASTM-B429	
(b)	Cotter bolts/ Hangers, Shackles, Brackets	Mild Steel	Hot dip galvanized	As per IS- 226 or IS-2062	
(c)	U Bolts	Stainless Steel or High Strength Al alloy 6061/ 65032	Forged && Heat treated	AISI 302 or 304-L ASTM- B429	
(d)	P. A. Rod	High Strength Al. Alloy 4600/ LM-6 or 6061/65032	Heat treatment during manufacturing	ASTM- B429	Min. tensile strength of 35 kg/mm2
3.	For AGS type clamp				
(a)	Supporting House	High Strength Corrosion resistant Al. Alloy 4600/ LM-6 or 6061/65032	Casted or forged & Heat treated	IS:617or ASTM-B429	
(b)	Al insert & Retaining strap	High Strength Al. Alloy 4600/ LM-6 or 6061/65032	Casted or forged & Heat treated	IS:617or ASTM-B429	High Strength Al. Alloy 4600/ LM-6 or 6061/ 65032
(c)	Elastomer	Moulded on Al. reinforcement			
4.	For Dead End Assembly				
(a)	Outer Sleeve	EC grade Al of purity not less than 99.50%			
(b)	~~Steel Sleeve~~	~~Mild Steel~~	~~Hot Dip Galvanized~~	~~IS:226/IS-2062~~	
5.	Ball & Socket Fittings,	Class-IV Steel	Drop forged & normalized Hot dip galvanized	As per IS: 2004	
6.	Yoke Plate	Mild Steel	Hot dip galvanized	As per IS- 226 or IS-2062	
7.	Sag Adjustment plate	Mild Steel	Hot dip galvanized	As per IS- 226 or IS-2062	
8(a)	Corona Control ring/ Grading ring	High Strength Al. Alloy tube (6061/ 6063/1100 type or 65032/ 63400 Type)	Heat treated Hot dip galvanized	ASTM-B429 or as per IS	Mechanical strength of welded joint shall not be less than 20 kN
8(b)	Supporting Brackets & Mounting Bolts	High Strength Al Alloy 7061/ 6063/ 65032/63400 Type) or Mild Steel	Heat treated Hot dip galvanized	ASTM-B429 or as per IS:226 or IS:2062	

Note : Alternate materials conforming to other national standards of other countries also may be offered provided the properties and compositions of these are close to the properties and compositions of material specified. Bidder should furnish the details of comparison of material offered viz a viz specified in the bid or else the bids are liable to be rejected.

2.0 Accessories for the AAAC Panther Conductor

2.1 General

2.1.1 This portion details the technical particulars of the accessories for Conductor.

2.1.2 2.5% extra fasteners, filler plugs and retaining rods shall be provided.

2.1.3 The supplier shall be responsible for satisfactory performance of complete conductor system along with accessories offered by him for continuous operation at temperature specified for the AAAC Panther Conductor.

2.2 Mid Span Compression Joint

2.2.1 Mid Span Compression Joint shall be used for joining two lengths of conductor. The joint shall have a resistively less than 75% of the resistivity of equivalent length of conductor. The joint shall not permit slipping off, damage to or failure of the complete conductor or any part thereof at a load less than 95% of the ultimate tensile strength of the conductor. It must be able to withstand the continuous design temperature of conductor.

2.2.2 The dimensions of mid span compression joint before & after compression along with tolerances shall be guaranteed in the relevant schedules of the bid and shall be decided by the manufacturer so as to suit the conductor size & conform to electrical & mechanical requirement stipulated in the specification. For composite core conductor, suitable sleeve, collets, collet housing shall be used for core jointing.

2.3 Repair Sleeve

Repair Sleeve of compression type shall be used to repair conductor with not more than two strands broken in the outer layer. The sleeve shall be manufactured from 99.5% pure aluminium / aluminium alloy and shall have a smooth surface. It shall be able to withstand the designed maximum operating temperature of conductor. The repair sleeve shall comprise of two pieces with a provision of seat for sliding of the keeper piece. The edges of the seat as well as the keeper piece shall be so rounded that the conductor strands are not damaged during installation. The dimensions of Repair sleeve along with tolerances shall be guaranteed in the relevant schedules of the bid and shall be decided by the manufacturer so as to suit the conductor size & conform to electrical & mechanical requirement stipulated in the specification.

2.4 Vibration Damper

2.4.1 Vibration dampers of 4R-stockbridge type with four (4) different resonances spread within the specified Aeolian frequency band width corresponding to wind speed of 1 m/s to 7 m/s are installed in the existing line at suspension and tension points on each conductor in each span along with bundle spacers to damp out Aeolian vibration as well as sub- span oscillations. One damper minimum on each side conductor / sub-conductor in a bundle for suspension points and one / two dampers minimum on each side conductor / sub-conductor in a bundle for tension points has been used for a ruling design span of --- meters.

2.4.2 The bidder shall offer damping system including Stockbridge type dampers and bundle spacers for AAAC Panther conductor for its protection from wind induced vibrations which could cause conductor fatigue/strand breakage near a hardware attachment, such as suspension clamps.

Alternate damping systems with proven design offering equivalent or better performance also shall be accepted provided the manufacturer meets the qualifying requirements stipulated in the Specifications. Relevant technical documents including type test reports to establish the technical suitability of alternate systems shall be furnished by the Bidder along with the bid.

The damper shall be designed to have minimum 4 nos of resonance frequencies to facilitate dissipation of vibration energy through inter-strand friction of the messenger cable and shall

be effective in reducing vibration over a wide frequency range (depending upon conductor dia) or wind velocity range specified above. The vibration damper shall meet the requirement of frequency or wind velocity range and also have mechanical impedance closely matched with the offered AAAC Panther conductor. The vibration dampers shall be installed at suitable positions to ensure damping effectiveness across the frequency range. The power dissipation of the vibration dampers shall exceed the wind power so that the vibration level on the conductor is reduced below its endurance limit i.e. 150 micro strain. The bidder shall clearly indicate the method for evaluating performance of dampers including analytical and laboratory test methods. The bidder shall indicate the type tests to evaluate the performance of offered damping system.

2.4.3 The clamp of the vibration damper shall be made of high strength aluminium alloy of type LM-6. It shall be capable of supporting the damper and prevent damage or chafing of the conductor during erection or continued operation. The clamp shall have smooth and permanent grip to keep the damper in position on the conductor without damaging the strands or causing premature fatigue failure of the conductor under the clamp. The clamp groove shall be in uniform contact with the conductor over the entire clamping surface except for the rounded edges. The groove of the clamp body and clamp cap shall be smooth, free from projections, grit or other materials which could cause damage to the conductor when the clamp is installed. Clamping bolts shall be provided with self- locking nuts and designed to prevent corrosion of threads or loosening in service.

2.4.4 The messenger cable shall be made of high strength galvanized steel/stain less steel with a minimum strength of 135 kg/sqmm. It shall be of preformed and post formed quality in order to prevent subsequent drop of weight and to maintain consistent flexural stiffness of the cable in service. The number of strands in the messenger cable shall be 19. The messenger cable other than stainless steel shall be hot dip galvanized in accordance with the recommendations of IS:4826 for heavily coated wires.

2.4.5 The damper mass shall be made of hot dip galvanized mild steel/cast iron or a permanent mould cast zinc alloy. All castings shall be free from defects such as cracks, shrinkage, inclusions and blowholes etc. The surface of the damper masses shall be smooth.

2.4.6 The damper clamp shall be casted over the messenger cable and offer sufficient and permanent grip on it. The messenger cable shall not slip out of the grip at a load less than the mass pull-off value of the damper. The damper masses made of material other-than zinc alloy shall be fixed to the messenger cable in a suitable manner in order to avoid excessive stress concentration on the messenger cables which shall cause premature fatigue failure of the same. The messenger cable ends shall be suitably and effectively sealed to prevent corrosion. The damper mass made of zinc alloy shall be casted over the messenger cable and have sufficient and permanent grip on the messenger cable under all service conditions.

2.4.7 The damper assembly shall be so designed that it shall not introduce radio interference beyond acceptable limits.

2.4.8 The vibration damper shall be capable of being installed and removed from energized line by means of hot line technique. in addition, the clamp shall be capable of being removed and reinstalled on the conductor at the designated torque without shearing or damaging of fasteners.

2.4.9 The Supplier must indicate the clamp bolt tightening torque to ensure that the slip strength of the clamp is maintained between 2.5kN and 5 kN. The clamp when installed on the conductor shall not cause excessive stress concentration on the conductor leading to permanent deformation of the conductor strands and premature fatigue failure in operation.

2.4.10 The vibration analysis of the system, with and without damper and dynamic characteristics of the damper as detailed under Appendix- I, shall have to be submitted. The technical particulars for vibration analysis and damping design of the system are as follows:

Sl.No.	Description	Technical particulars
1. i) ii) iii)	Span length in meters Ruling design span Maximum span Minimum span	 ---meters (e.g. 320m) 1100 meters 100 meters
2.	Configuration	----[e.g. Double Circuit twin bundle conductor per phase in vertical configuration]
3.	Tensile load in Conductor at temperature of minimum temperature and still air	As per Sag – tension calculations
4.	Armour rods used	Standard preformed armour rods/AGS
5.	Maximum permissible dynamic strain i.e. endurance limit.	+/- 150 micro strains

2.4.11 The damper placement chart shall be submitted for spans ranging from 100m to 1100m. Placement charts should be duly supported with relevant technical documents and sample calculations.

2.4.15 The damper placement charts shall include the following

(1) Location of the dampers for various combinations of spans and line tensions clearly indicating the number of dampers to be installed per conductor per span.

(2) Placement distances clearly identifying the extremities between which the distances are to be measured.

(3) Placement recommendation depending upon type of suspension clamps (viz Free center type/Armour grip type etc.)

(4) The influence of mid span compression joints, repair sleeves and armour rods (standard and AGS) in the placement of dampers.

2.5 Deleted

2.6 Deleted

2.7 Material and Workmanship

2.7.1 All the equipment shall be of the latest proven design and conform to the best modern practice adopted in the extra high voltage field. The Bidder shall offer only such equipment as guaranteed by him to be satisfactory and suitable for 132kV transmission line application will give continued good performance at all service conditions. For employer's review of the offered design of accessories, the supplier shall submit document/design details of similar type of accessories used in past for similar type of AAAC Panther conductor application

2.7.2 The design, manufacturing process and quality control of all the materials shall be such as to achieve requisite factor of safety for maximum working load, highest mobility, elimination of sharp edges and corners, best resistance to corrosion and a good finish.

2.7.3 High current, heat rise test shall be conducted by the supplier to determine the maximum temperature achieved in different components of fittings/ accessories under simulated service condition corresponding to continuous operation of conductor at designed maximum temperature. The material of the components should be suitable for continued good performance corresponding to these maximum temperatures. The supplier shall submit relevant type/ performance test certificates as per applicable standards/product specifications to confirm suitability of the offered material.

2.7.4 All ferrous parts shall be hot dip galvanized, after all machining has been completed. Nuts may, however, be tapped (threaded) after galvanizing and the threads oiled. Spring washers shall be electro galvanized as per grade 4 of IS-1573. The bolt threads shall be undercut to take care of increase in diameter due to galvanizing. Galvanizing shall be done in accordance

with IS:2629/ IS-1367 (Part-13) and satisfy the tests mentioned in IS-2633. Fasteners shall withstand four dips while spring washers shall withstand three dips. Other galvanized materials shall have a minimum average coating of Zinc equivalent to 600 gm/sq.m and shall be guaranteed to withstand at least six dips each lasting one minute under the standard Preece test for galvanizing unless otherwise specified.

2.7.5 The zinc coating shall be perfectly adherent, of uniform thickness, smooth, reasonably bright, continuous and free from imperfections such as flux, ash, rust stains, bulky white deposits and blisters. The zinc used for galvanizing shall be of grade Zn 99.95 as per IS:209.

2.7.6 In case of castings, the same shall be free from all internal defects like shrinkage, inclusion, blow holes, cracks etc.

2.7.7 All current carrying parts shall be so designed and manufactured that contact resistance is reduced to minimum and localized heating phenomenon is averted.

2.7.8 No equipment shall have sharp ends or edges, abrasions or projections and shall not cause any damage to the conductor in any way during erection or during continuous operation which would produce high electrical and mechanical stresses in normal working. The design of adjacent metal parts and mating surfaces shall be such as to prevent corrosion of the contact surface and to maintain good electrical contact under all service conditions.

2.7.9 Particular care shall be taken during manufacture and subsequent handling to ensure smooth surface free from abrasion or cuts.

2.7.10 The fasteners shall conform to the requirements of IS:6639-1972. All fasteners and clamps shall have corona free locking arrangement to guard against vibration loosening.

2.8 Compression Markings

Die compression areas shall be clearly marked on each equipment designed for continuous die compressions and shall bear the words 'COMPRESS FIRST' 'suitably inscribed on each equipment where the compression begins. If the equipment is designed for intermittent die compressions, it shall bear the identification marks 'COMPRESSION ZONE' and 'NON-COMPRESSION ZONE' distinctly with arrow marks showing the direction of compression and knurling marks showing the end of the zones. The letters, number and other markings on finished equipment shall be distinct and legible.

3.0 Tests and Standards

3.1 Type Tests

3.1.1 On Suspension Clamp

a) Magnetic power loss test : As per Appendix-I

b) Clamp slip strength Vs torque test : As per Appendix-I

c) Ozone Test on elastomer : As per Appendix-I

d) Vertical damage load & Failure load test : IEC:61284

3.1.2 On Dead end Tension Assembly

a) Electrical resistance test for dead end Assembly : As per IS:2486- (Part-I)

b) Heating cycle test for dead end Assembly : As per Appendix-I

c) Slip strength test for dead end assembly : As per Appendix-I

d) Ageing test on filler (if applicable) : As per Appendix-I

3.1.3 Mid Span Compression Joint for Conductor

a) Chemical analysis of materials : As per Appendix-I

b) Electrical resistance test :As per IS:2121 (Part-II)

c) Heating cycle test : As per Appendix-I

d) Slip strength test : As per Appendix-I

e) Corona extinction voltage test (dry)[for 400kV] : As per Appendix-I

f) Radio interference voltage test (dry) [for 400kV]: As per Appendix-I

3.1.4 Repair Sleeve for Conductor

a) Chemical analysis of materials : As per Appendix-I

b) Corona extinction voltage test (dry) [for 400kV] : As per Appendix-I

c) Radio interference voltage test (dry) [for 400kV]: As per Appendix-I

3.1.5 Vibration Damper for Conductor

a) Chemical analysis of materials : As per Appendix-I

b) Dynamic characteristics test* : As per Appendix-I

c) Vibration analysis : As per Appendix-I

d) Clamp slip test : As per Appendix-I

e) Fatigue tests : As per Appendix-I

f) Magnetic power loss test : As per Appendix-I

g) Corona extinction voltage test (dry) [for 400kV] : As per Appendix-I

h) Radio interference voltage test (dry) [for 400kV]: As per Appendix-I

i) Damper efficiency test : As per IS:9708

* Applicable for 4 R Stockbridge dampers. For alternate type of vibration dampers (permitted as per clause 2.4.2), as an alternative to dynamic characteristic test, damper efficiency test as per IEEE-664 may be proposed/ carried out by the supplier.

Type tests specified above shall not be required to be carried out if a valid test certificate is available for a same design, i.e., tests conducted earlier should have been conducted in accredited laboratory (accredited based on ISO/IEC guide 25/17025 or EN 45001 by the National Accreditation body of the country where laboratory is located) or witnessed by the representative (s) of a Utility.

In the event of any discrepancy in the test report (i.e., any test report not applicable due to any design / material/manufacturing process change including substitution of components or due to non-compliance with the requirement stipulated in the Technical Specification) the tests shall be conducted by the Supplier at no extra cost to the Purchaser.

3.2 Acceptance Tests

3.2.1 On Both Suspension Clamp and Tension Assembly

a) Visual Examination : As per IS:2486-(Part-I)

b) Verification of dimensions : As per IS:2486-(Part-I)

c) Galvanizing/Electroplating test : As per IS:2486-(Part-I)

d) Mechanical strength test of each component : As per Appendix-I

e) Mechanical Strength test of welded joint : As per Appendix-I

f) Chemical analysis, hardness tests, grain size, inclusion rating & magnetic particle inspection for forgings/castings : As per Appendix-I

3.2.2 On Suspension Clamp only

a) Clamp Slip strength Vs Torque test for suspension clamp : As per Appendix-I

b) Shore hardness test of elastomer cushion for AG suspension clamp : As per Appendix-I

c) Bend test for armour rod set : As per IS:2121(Part-I), Clause 7.5,7,10 & 7.11

d) Resilience test for armour rod set : As per IS:2121(Part-I), Clause 7.5,7,10 & 7.11

e) Conductivity test for armour rods set : As per IS:2121(Part-I), Clause 7.5,7,10 & 7.11

3.2.3 On Tension Hardware Fittings only

a) Slip strength test for dead end assembly : As per Appendix-I

d) Ageing test on filler (if applicable) : As per Appendix-II

3.2.4 On Mid Span Compression Joint for Conductor

a) Visual examination and dimensional verification : As per IS:2121 (Part-II), Clause 6.2, 6.3

b) Galvanizing test : As per Appendix-II

c) Hardness test : As per Appendix-II

d) Ageing test on filler (if applicable) : As per Appendix-I

3.2.5 Repair Sleeve for Conductor

a) Visual examination and dimensional verification : As per IS:2121(Part-II) Clause 6.2, 6.3

3.2.6 Vibration Damper for Conductor

a) Visual examination and dimensional verification : As per IS:2121(Part-II) Clause 6.2, 6.3

b) Galvanizing test : As per Appendix-II

(i) On damper masses : As per Appendix-II

ii) On messenger cable

c) Verification of resonance frequencies : As per Appendix-II

d) Clamp slip test : As per Appendix-II

e) Clamp bolt torque test : As per Appendix-II

f) Strength of the messenger cable : As per Appendix-II

g) Mass pull off test : As per Appendix-II

h) Dynamic characteristics test* : As per Appendix-II

* Applicable for 4 R Stockbridge dampers. For alternate type of vibration dampers (permitted as per clause 2.4.2), as an alternative to dynamic characteristic test, damper efficiency test as per IEEE-664 may be proposed/ carried out by the supplier.

3.3 Routine Tests

3.3.1 For Hardware Fittings

a) Visual examination IS:2486-(Part-I)

b) Proof Load Test : As per Appendix-I

3.3.2 For conductor accessories

a) Visual examination and dimensional verification : As per IS:2121(Part-II) Clause 6.2, 6.3

3.4 Tests during Manufacture on all components as applicable

a) Chemical analysis of Zinc used for galvanizing IS:2486-(Part-I)

b) Chemical analysis mechanical metallographic test and magnetic particle inspection for malleable castings : As per Appendix-I

c) Chemical analysis, hardness tests and magnetic particle inspection for forging : As per Appendix-I

3.5 Testing Expenses

3.5.1 Testing charges for the type test specified shall be indicated separately in the prescribed schedule.

3.5.2 Bidder shall indicate charges for all type tests covered under Clause No. 3.1.1 to 3.1.8 separately. The charges for each type test shall be separately indicated.

3.5.6 Bidder shall indicate the laboratories in which they propose to conduct the type tests. They shall ensure that adequate facilities for conducting the tests are available in the laboratory and the tests can be completed in these laboratories within the time schedule guaranteed by them in the appropriate schedule.

3.5.7 The entire cost of testing for acceptance and routine tests and tests during manufacture specified herein shall be treated as included in the quoted Ex-works/CIF Price.

3.5.8 In case of failure in any type test, repeat type tests are required to be conducted, then, all the expenses for deputation of Inspector/ Purchaser's representative shall be deducted from the contract price. Also if on receipt of the Supplier's notice of testing, the Purchaser's representative/Inspector does not find material & facilities to be ready for testing the expenses incurred by the Purchaser's for re- deputation shall be deducted from contract price.

3.5.9 The Supplier shall intimate the Purchaser about carrying out of the type tests along with detailed testing program at least 3 weeks in advance (in case of testing in India and at least 6 weeks advance in case of testing abroad) of the scheduled date of testing during which the Purchaser will arrange to depute his representative to be present at the time of carrying out the tests.

3.6 Sample Batch For Type Testing

3.6.1 The Supplier shall offer material for sample selection for type testing only after getting Quality Assurance Programme approved by the Purchaser. The Supplier shall offer at least three times the quantity of materials required for conducting all the type tests for sample selection. The sample for type testing will be manufactured strictly in accordance with the Quality Assurance Programme approved by the Purchaser.

3.6.2 Before sample selection for type testing the Supplier shall be required to conduct all the acceptance tests successfully in presence of Purchaser's representative.

3.7 Schedule of Testing and Additional Tests

3.7.1 The Bidder has to indicate the schedule of following activities in their bids

(a) Submission of drawing for approval.

(b) Submission of Quality Assurance programme for approval.

(c) Offering of material for sample selection for type tests.

(d) Type testing.

3.7.2 The Purchaser reserves the right of having at his own expense any other test(s) of reasonable nature carried out at Supplier's premises, at site, or in any other place in addition to the aforesaid type, acceptance and routine tests to satisfy himself that the material comply with the specifications.

3.7.3 The Purchaser also reserves the right to conduct all the tests mentioned in this specification at his own expense on the samples drawn from the site at Supplier's premises or at any other test centre. In case of evidence of non-compliance, it shall be binding on the part of Supplier to prove the compliance of the items to the technical specifications by repeat tests, or correction of deficiencies, or replacement of defective items, all without any extra cost to the Purchaser.

3.8 Test Reports

3.8.1 Copies of type test reports shall be furnished in at least six copies along with one original. One copy shall be returned duly certified by the Purchaser , only after which the commercial production of the concerned material shall start.

3.8.2 Copies of acceptance test report shall be furnished in at least six copies. One copy shall be returned, duly certified by the Purchaser, only after which the materials will be dispatched.

3.8.3 Record of routine test report shall be maintained by the Supplier at his works for periodic inspection by the Purchaser's representative.

3.8.4 Test certificates of tests during manufacture shall be maintained by the Supplier. These shall be produced for verification as and when desired by the Purchaser.

3.9 Inspection

3.9.1 The Purchaser's representative shall at all times be entitled to have access to the works and all places of manufacture, where the material and/or its component parts shall be manufactured and the representatives shall have full facilities for unrestricted inspection of the Supplier's, sub-Supplier's works raw materials. manufacturers of all the material and for conducting necessary tests as detailed herein.

3.9.2 The material for final inspection shall be offered by the Supplier only under packed condition as detailed in clause 3.10 of this part of the Specification. The engineer shall select samples at random from the packed lot for carrying out acceptance tests.

3.9.3 The Supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of material in its various stages so that arrangements could be made for inspection.

3.9.4 Material shall not be dispatched from its point of manufacture before it has been satisfactorily inspected and tested unless the inspection is waived off by the Purchaser in writing. In the latter case also the material shall be dispatched only after all tests specified herein have been satisfactorily completed.

3.9.5 The acceptance of any quantity of material shall in no way relieve the Supplier of his responsibility for meeting all the requirements of the Specification, and shall not prevent subsequent rejection, if such material are later found to be defective.

3.10 Packing and Marking

3.10.1 All material shall be packed in strong and weather resistant wooden cases/crates. The gross weight of the packing shall not normally exceed 200 Kg to avoid handling problems.

3.10.2 The packing shall be of sufficient strength to withstand rough handling during transit, storage at site and subsequent handling in the field.

3.10.3 Suitable cushioning, protective padding, dunnage or spacers shall be provided to prevent damage or deformation during transit and handling.

3.10.4 Bolts, nuts, washers, cotter pins, security clips and split pins etc. shall be packed duly installed and assembled with the respective parts and suitable measures shall be used to prevent their loss.

3.10.5 Each component part shall be legibly and indelibly marked with trade mark of the manufacturer and year of manufacture.

3.10.6 All the packing cases shall be marked legibly and correctly so as to ensure safe arrival at their destination and to avoid the possibility of goods being lost or wrongly dispatched on account of faulty packing and faulty or illegible markings. Each wooden case/crate shall have all the markings stencilled on it in indelible ink.

3.11 Standards

3.11.1 The Hardware fittings; conductor and earth wire accessories shall conform to the following Indian/International Standards which shall mean latest revisions, with amendments/changes adopted and published, unless specifically stated otherwise in the Specification.

3.11.2 In the event of the supply of hardware fittings; conductor and earthwire accessories conforming to standards other than specified, the Bidder shall confirm in his bid that these standards are equivalent to those specified. In case of award, salient features of comparison between the Standards proposed by the Supplier and those specified in this document will be provided by the Supplier to establish their equivalence.

Sl.No.	Indian Standard	Title	International Standard
1.	IS: 209-1992	Specification for zinc	BS:3436-1986
2.	IS:398	Aluminum Conductor Galvanized Steel- Reinforced For Extra High Voltage	IEC:1089-1991 BS:215-1970
3.	IS 1573	Electroplated Coating of Zinc on iron and Steel	
4.	IS : 2121 (Part-II)	Specification for Conductor and Earthwire Accessories for Overhead Power lines: Mid-span Joints and Repair Sleeves for Conductors	
5.	IS:2486 (Part-I)	Specification for Insulator Fittings for Overhead power Lines with Nominal Voltage greater than 1000 V: General Requirements and Tests	
6.	IS:2629	Recommended Practice for Hot Dip Galvanizing of Iron and Steel	
7.	IS:2633	Method of Testing Uniformity of Coating on Zinc Coated Articles	
8.		Ozone test on Elastomer	ASTM- D1 171
9.		Tests on insulators of Ceramic material or glass for overhead lines with a nominal voltage greater than 1000V	IEC:383-1993
10.	IS:4826	Galvanized Coating on Round Steel Wires	ASTM A472- 729 BS:443-1969
11.	IS:6745	Methods of Determination of Weight of Zinc Coating of Zinc Coated Iron and Steel Articles	BS:433 ISO : 1460 (E)
12.	IS:8263	Method of Radio Interference Tests on High Voltage Insulators	IEC:437 NEMA:107 CISPR
13.	IS:6639	Hexagonal Bolts for Steel Structures	ISO/R-272

14.	IS:9708	Specification for Stock Bridge Vibration Dampers for Overhead Power Lines	
15.	IS:10162	Specification for Spacers Dampers for Twin Horizontal Bundle Conductors	

1.0 Tests on Hardware Fittings

1.1 Magnetic Power Loss Test for Suspension Assembly

Two hollow aluminium tubes of 32 mm diameter for the conductor shall be placed 450 mm (for 400kV) apart. An alternating current over the range of 1200 to 1800 amps shall be passed through each tube. One hollow aluminium tubes of 29mm diameter for the conductor shall be used for 132kV. An alternating current over the range of 300 - 700 amps shall be passed through the tube. The reading of the watt meter with and without suspension assemblies along with line side yoke plate, clevis eye shall be recorded. Not less than three suspension assemblies shall be tested. The average power loss for suspension assembly shall be plotted for each value of current. The value of the loss corresponding to ---A (at steady state conductor temperature) shall be read off from the graph and the same shall be limited to the value guaranteed by the supplier.

1.2 Galvanizing/Electroplating Test

The test shall be carried out as per Clause no. 9.4 of IS: 2486-(Part-1) except that both uniformity of zinc coating and standard preece test shall be carried out and the results obtained shall satisfy the requirements of this specification.

1.3 Mechanical Strength Test of Each Component

Each component shall be subjected to a load equal to the specified minimum ultimate tensile strength (UTS) which shall be increased at a steady rate to 67% of the minimum UTS specified. The load shall be held for five minutes and then removed. The component shall then again be loaded to 50% of UTS and the load shall be further increased at a steady rate till the specified UTS and held for one minute. No fracture should occur. The applied load shall then be increased until the failing load is reached and the value recorded.

1.4 Mechanical Strength Test of Welded Joint

The welded portion of the component shall be subjected to a Load of 2000 kgs for one minute. Thereafter, it shall be subjected to die-penetration/ ultrasonic test. There shall not be any crack at the welded portion.

1.5 Clamp Slip Strength Vs Torque Test for Suspension Clamp

The suspension assembly shall be vertically suspended by means of a flexible attachment. A suitable length of conductor shall be fixed in the clamp. The clamp slip strength at various tightening torques shall be obtained by gradually applying the load at one end of the conductor. The Clamp slip strength vs torque curve shall be drawn. The above procedure is applicable only for free center type suspension clamp. For AG suspension clamp only clamp slip strength after assembly shall be found out. The clamp slip strength at the recommended tightening torque shall be as indicated in the GTP. [e.g. for 400kV: more than 20 kN but less than 29 kN.

1.6 Heating Cycle Test

Heating cycle test shall be performed in accordance with IS 2486 (Part-I) with following modifications:

i) Temperature of conductor during each cycle: 40 deg. C above designed maximum operating temperature of the conductor, but not exceeding the emergency temperature of the conductor.

ii) Number of cycle: 100

iii) Slip strength test shall also be carried out after heating cycle test.

1.7 Slip strength test for dead end assembly

The test shall be carried out as per IS:2486 (Part-I) except that the load shall be steadily increased to 95% of minimum ultimate tensile strength of conductor/earth wire and retained for one minute at this load.

1.8 Ageing Test on Filler (if applicable)

The test shall be done in accordance with Grease drop point test method. The specimen should be drop as a droplet when kept at a temperature 40 deg. C above designed maximum operating temperature of the conductor for 30 minutes. The temperature shall then be increase till one droplet drops and the temperature recorded.

1.9 Shore Hardness Test for Elastomer Cushion for AG Suspension Assembly

The shore hardness at various points on the surface of the elastomer cushion shall be measured by a shore hardness meter and the shore hardness number shall be between 65 to 80.

1.10 Proof Load Test

Each component shall be subjected to a load equal to 50% of the specified minimum ultimate tensile strength which shall be increased at a steady rate to 67% of the UTS specified. The load shall be held for one minute and then removed. After removal of the load the component shall not show any visual deformation.

1.11 Tests for Forging Casting and Fabricated Hardware

The chemical analysis, hardness test, grain size, inclusion rating and magnetic particle inspection for forging, castings and chemical analysis and proof load test for fabricated hardware shall be as per the internationally recognized procedures for these tests. The sampling will be based on heat number and heat treatment batch. The details regarding test will be as in the Quality Assurance programme.

1.12 Ozone Test for Elastomer

This test shall be performed in accordance with ASTM D-1171 by the Ozone chamber exposure method (method B). The test duration shall be 500 hours and the ozone concentration 50 PPHM. At the test completion, there shall be no visible crack under a 2 x magnification.

2.0 Tests on Accessories for Conductor

2.1 Mid Span Compression Joint for Conductor

(a) Slip Strength Test

The fitting compressed on conductor shall not be less than one metre in length. The test shall be carried out as per IS:2121 (Part-ii)-1981 clause 6.4 except that the load shall be steadily increased to 95% of minimum ultimate tensile strength of conductor/earthwire and retained for one minute at this load. There shall be no movement of the conductor/ earthwire relative to the fittings and no failure of the fittings during this one minute period.

(b) Heating Cycle Test

Heating cycle test shall be performed in accordance with IS 2121 (Part- II- 1981) with following modifications:-

i) Temperature of conductor during each cycle: 40 deg. C above designed maximum operating temperature of the conductor.

ii) Number of cycle: 100

iii) Slip strength test shall also be carried out after heating cycle test.

2.2 Vibration Damper for Conductor

(a) Dynamic Characteristics Test

The damper shall be mounted with its clamp tightened with torque recommended by the manufacturer on shaker table capable of simulating sinusoidal vibrations for Aeolian vibration frequency band ranging from 0.18/d to 1.4/d where d is the conductor diameter in meters. The damper assembly shall be vibrated vertically with a + 1 mm amplitude from 5 to 15 Hz frequency and beyond 15 Hz at ± 0.5mm to determine following characteristics with the help of suitable recording instruments:

(i) Force Vs frequency

(ii) Phase angle Vs frequency

(iii) Power dissipation Vs frequency

The Force Vs frequency curve shall not show steep peaks at resonance frequencies and deep troughs between the resonance frequencies. The resonance frequencies shall be suitably spread within the Aeolian vibration frequency-band between the lower and upper dangerous frequency, limits determined by the vibration analysis of conductor without dampers.

Acceptance criteria for vibration damper:

(i) The above dynamic characteristics test on five damper shall be conducted.

(ii) The mean reactance and phase angle Vs frequency curves shall be drawn with the criteria of best fit method.

(iii) The above mean reactance response curve should lie within 0.191 f to 0.762 f Kgf/mm limits where f is frequency in Hz.

(iv) The above mean phase angle response curve shall be between 25^0 to 130^0 within the frequency range of interest.

(v) If the above curve lies within the envelope, the damper design shall be considered to have successfully met the requirement.

(vi) Visual resonance frequencies of each mass of damper is to be recorded and to be compared with the guaranteed values.

(b) Vibration Analysis

The vibration analysis of the conductor shall be done with and without damper installed on the span. The vibration analysis shall be done on a digital computer using energy balance approach.

The following parameters shall be taken into account for the purpose of analysis:

(i) The analysis shall be done for single conductor without armour rods as per the parameters given in the Specification. The tension shall be taken from Sag & Tension calculation (0 deg. C & no wind condition and m ruling span) for a span ranging from 100 m to 1100 m.

(ii) The self-damping factor and flexural stiffness (El) for conductor shall be calculated on the basis of experimental results. The details for experimental analysis with these data should be furnished.

(iii) The power dissipation curve obtained from Dynamic Characteristics Test shall be used for analysis with damper.

(iv) Examine the aeolian vibration level of the conductor with and without vibration damper installed at the recommended location or wind velocity ranging from 0 to 30 Km per hour, predicting amplitude, frequency and vibration energy input.

(v) From vibration analysis of conductor without damper, anti-node vibration amplitude and dynamic strain levels at clamped span extremities as well as antinodes shall be examined and thus lower and upper dangerous frequency limits between which the Aeolian vibration levels exceed the specified limits shall be determined.

(vi) From vibration analysis of conductor with damper/dampers installed at the recommended location, the dynamic strain level, at the clamped span extremities, damper attachment point and the antinodes on the conductor shall be determined. In addition to above damper clamp vibration amplitude and anti-node vibration amplitudes shall also be examined.

The dynamic strain levels at damper attachment points, clamped span extremities and antinodes shall not exceed the specified limits. The damper clamp vibration amplitude shall not be more than that of the specified fatigue limits.

(c) Clamp Slip and Fatigue Tests

(i) Test Set Up

The clamp slip and fatigue tests shall be conducted on a laboratory set up with a minimum effective span length of 30 m. The conductor shall be tensioned at tension corresponding to minimum temperature & no wind condition and ruling span ----m from sag –tension calculation and shall not be equipped with protective armour rods at any point.

Constant tension shall be maintained within the span by means of lever arm arrangement. After the conductor has been tensioned, clamps shall be installed to support the conductor at both ends and thus influence of connecting hardware fittings are eliminated from the free span. The clamps shall not be used for holding the tension on the conductor. There shall be no loose parts, such as suspension clamps, U bolts on the test span supported between clamps mentioned above. The span shall be equipped with vibration inducing equipment suitable for producing steady standing vibration. The inducing equipment shall have facilities for stepless speed control as well as stepless amplitude arrangement. Equipment shall be available for measuring the frequency, cumulative number of cycles and amplitude of vibration at any point along the span.

(ii) Clamp Slip test

The vibration damper shall be installed on the test span. The damper clamp, after tightening with the manufacturer's specified tightening torque, when subjected to a longitudinal pull of 2.5 kN parallel to the axis of conductor for a minimum duration of one minute shall not slip i.e. the permanent displacement between conductor and clamp measured after removal of the load shall not exceed 1.0 mm. The load shall be further increased till the clamp starts slipping. The load at which the clamp slips shall not be more than 5 kN.

(iii) Fatigue Test

The vibration damper shall be installed on the test span with the manufacturer's specified tightening torque. It shall be ensured that the damper shall be kept minimum three loops away from the shaker to eliminate stray signals influencing damper movement.

The damper shall then be vibrated at the highest resonant frequency of each damper mass. For dampers involving resonant frequencies, tests shall be done at torsional modes also in addition to the highest resonant frequencies at vertical modes. The resonance frequency shall be identified as the frequency at which each damper mass vibrates with the maximum amplitude

on itself. The amplitude of vibration of the damper clamp shall be maintained not less than ± 25/f mm, where f is the frequency in Hz.

The test shall be conducted for minimum ten million cycles at each resonant frequency mentioned above. During the, test if resonance shift is observed the test frequency shall be tuned to the new resonant frequency.

The clamp slip test as mentioned hereinabove shall be repeated after fatigue test without re-torquing or adjusting the damper clamp, and the clamp shall withstand a minimum load equal to 80% of the slip strength for a minimum duration of one minute.

After the above tests, the damper shall be removed from conductor and subjected to dynamic characteristics test. There shall not be any major deterioration in the characteristic of the damper. The damper then shall be cut open and inspected. There shall not be any broken, loose, or damaged part. There shall not be significant deterioration or wear of the damper. The conductor under clamp shall also be free from any damage.

For the purpose of acceptance, the following criteria shall be applied.

I. There shall not be any frequency shift by more than +2 Hz for frequencies lower than 15 Hz and ± 3 Hz for frequencies higher than 15 Hz.

II. The force response curve shall generally lie within guar anteed % variation in reactance after fatigue test in comparison with that before fatigue test by the Supplier.

III. The power dissipation of the damper shall not be less than guaranteed % variation in power dissipation before fatigue test by the Supplier. However, it shall not be less than minimum power dissipation which shall be governed by lower limits of reactance and phase angle indicated in the envelope.

2.3 Deleted

2.4 Deleted

2.5 Corona Extinction Voltage Test (Dry) [for 400kV]

The sample when subjected to power frequency voltage shall have a corona extinction voltage of not less than 320 kV rms line to ground under dry condition. There shall be no evidence of corona on any part of the sample. The atmospheric condition during testing shall be recorded and the test results shall be accordingly corrected with suitable correction factor as stipulated in IS:731- 1971.

2.6 Radio Interference Voltage Test (Dry) [for 400kV]

Under the conditions as specified under (3.8) above, the sample shall have a radio interference voltage level below 1000 microvolts at one MHz when subjected to 50 Hz AC voltage of 305 kV rms line to ground under dry condition. The test procedure shall be in accordance with IS:8263.

2.7 Chemical Analysis Test

Chemical analysis of the material used for manufacture of items shall be conducted to check the conformity of the same with Technical Specification and approved drawing.

3.0 Tests on All components (As applicable)

3.1 Chemical Analysis of Zinc used for Galvanizing

Samples taken from the zinc ingot shall be chemically analysed as per IS- 209-1979. The purity of zinc shall not be less than 99.95%.

3.2 Tests for Forgings

The chemical analysis hardness tests and magnetic particle inspection for forgings, will be as per the internationally recognized procedures for these tests. The, sampling will be based on heat number and heat treatment batch. The details regarding test will be as discussed and mutually agreed to by the Supplier and Purchaser in Quality Assurance Programme.

3.3 Tests on Castings

The chemical analysis, mechanical and metallographic tests and magnetic particle inspection for castings will be as per the internationally recognized procedures for these tests. The samplings will be based on heat number and heat treatment batch. The details regarding test will be as discussed and mutually agreed to by the Supplier and Purchaser in Quality Assurance Programme.

Appendix-II

Acceptance Tests

1. Mid Span Compression Joint for Conductor

(a) Hardness Test

The Brinnel hardness at various points on the steel sleeve of conductor core and tension clamp shall be measured.

2. Vibration Damper for Conductor

(a) Verification of Resonance Frequencies

The damper shall be mounted on a shaker table and vibrate at damper clamp displacement of +/-0.5 mm to determine the resonance frequencies. The resonance shall be visually identified as the frequency at which damper mass vibrates with maximum displacement on itself. The resonance frequency thus identified shall be compared with the guaranteed value. A tolerance of ± 1 Hz at a frequency lower than 15 Hz and ± 2 Hz at a frequency higher than 15 Hz only shall be allowed.

(b) Clamp Slip Test

Same as Clause 2.2 (c) (ii) of Appendix-I.

(c) Clamp Bolt Torque Test

The clamp shall be attached to a section of the conductor/earthwire. A torque of 150 percent of the manufacturer's specified torque shall be applied to the bolt. There shall be no failure of component parts. The test set up is as described in Clause 2.2 (c) (i), Appendix-I.

(d) Strength of the Messenger Cable

The messenger cable shall be fixed in a suitable tensile testing machine and the tensile load shall be gradually applied until yield point is reached. Alternatively, each strand of messenger cable may be fixed in a suitable tensile testing machine and the tensile load shall be gradually applied until yield point is reached. In such a case, the 95% of yield strength of each wire shall be added to get the total strength of the cable. The load shall be not less than the value guaranteed by the Supplier.

(e) Mass Pull off Test

Each mass shall be pulled off in turn by fixing the mass in one jaw and the clamp in the other of a suitable tensile testing machine. The longitudinal pull shall be applied gradually until the mass begins to pull out of the messenger cable. The pull off loads shall not be less than the value guaranteed by the Supplier.

(f) Dynamic Characteristics Test

The test will be performed as acceptance test with the procedure mentioned for type test with sampling mentioned below:

Vibration Damper :

- 1 Sample for lot of 1000 Nos. & below
- 3 Samples for lot above 1000 & up to 5000 nos.
- Additional 1 sample for every additional 1500 pieces above 5000. The acceptance criteria will be as follows:

(i) The above dynamic characteristics curve for reactance & phase angle

will be done for frequency range of 5 Hz to 40 Hz.

(ii) If all the individual curve for dampers are within the envelope as already mentioned for type test for reactance & phase angle, the lot passes the test.

(iii) If individual results do not fall within the envelope, averaging of characteristics shall be done.

(iv) Force of each damper corresponding to particular frequency shall be taken & average force of three dampers at the frequency calculated.

(v) Similar averaging shall be done for phase angle.

(vi) Average force Vs frequency and average phase Vs frequency curves shall be plotted on graph paper. Curves of best fit shall be drawn for the entire frequency range.

(vii) The above curves shall be within the envelope specified.

(g) Clamp Slip Test

Same as clause 2.3(e) of Appendix-I.

(h) Clamp Bolt Torque Test

The spacer assembly shall be attached to conductor. A torque of 150 per cent of the manufacturer's specified tightening torque shall be applied to the clamp bolts or cap screws. There shall be no failure of the component parts.

(i) Assembly Torque Test

The spacer assembly shall be installed on conductor. The same shall not rotate on either clamp on applying a torque of 0.04 kN in clockwise or anti- clockwise direction.

(j) Hardness test for Elastomer

The shore hardness at different points on the elastomer surface of cushion grip clamp shall be measured by shore hardness meter. They shall lie between 65 to 80.

(k) UTS of Retaining Rods

The ultimate tensile strength of the retaining rods shall be measured. The value shall not be less than 35 kg/sq.mm.

(l) Ageing Test on filler (if applicable)

Same as clause 1.8 of Appendix-I.

Appendix-I Standard Technical Particulars of Hardware Fittings and Accessories

1. Suspension hardware fittings for AAAC Panther conductor

S.No.	Description	Unit	Particulars/ Value		
			Single "I" Suspension Fittings with		Single suspension Pilot Fitting with
			AGS clamp	Free Centre clamp	Envelope clamp
1.	Maximum magnetic power loss of one suspension assembly at sub-conductor current of 600 amperes	Watt	4		8
2.	Slipping strength of suspension assembly	KN	9-18		9-18
3.	Particulars of standard/ AGS preformed armour rod set for suspension assembly				
	a) No. of rods per set	No.	12		NA
	b) Direction of lay		Right hand		NA
	c) Overall length after fitting on conductor	mm	2235	2540	NA
	d) Diameter of each rod	mm	9.27		NA
	e) Tolerance in				
	i) Diameter of each rod	±mm	0.10		NA
	ii) Length of each rod	±mm	25		NA
	iii) Difference of length between the longest and shortest rod in a set	±mm	13		NA
	f) Type of Aluminium alloy used for manufacture of PA rod set		6061/ 65032		NA
	g) Minimum UTS of each rod	Kg/mm2	35		NA

2. Tension hardware fittings for AAAC PANTHER Conductor

S.No.	Description	Unit	Particulars/Value	
			Single Tension	Double Tension
1.	Mechanical strength of Tension fitting (excluding dead end clamp)	KN	120	2x120
2.	Type of dead-end assembly		Compression	
3.	Compression pressure	MT	100	
4.	Maximum electrical resistance of dead-end assembly as a percentage of equivalent length of Conductor	%	75	
5.	Slip strength of dead-end assembly	KN	64.1 kN (95% of conductor UTS-67.49kN)	
6.	Galvanising			
	a) Minimum weight of Zinc coating for steel parts	gm/m2	900	
	b) Purity of Zinc used for galvanising	%	99.95 (IS 209) or 98.5 (IS 13229)	
	c) Min. No. of dips in standard preece test the ferrous parts can withstand (wherever applicable)	No.	a) Fasteners: 4 dips of 1 minute b) Spring washers: 3 dips of 1 minute & c) all others: 6 dips of 1 minute	

5. Mid span compression Joint for AAAC PANTHER Conductor

S.No.	Description	Unit	Particulars/ Value
			Aluminium Alloy Sleeve-Grade 6061

1.	Material of Joint		Aluminium of purity 99.5% As per SS 4240813 & SS 4240814, IS-9997
2.	Range of Hardness of the steel sleeve (Brinnel hardness)	BHN	From 100 to 200
3.	Weight of Zinc coating for steel sleeve	gm/m2	900
4.	Dimension of sleeve Before compression		Aluminium Alloy Sleeve-Grade 6061
	i) Inside diameter	mm	23.00 ± 0.5
	ii) Outside diameter	mm	38.00 ± 1.0
	iii) Length	mm	610 ± 5
5.	Dimensions of Sleeve after compression		Aluminium Alloy Sleeve-Grade 6061
	i) Outside dimension (Corner to corner)	mm	37.00 ± 0.5
	ii) Outside dimension (face to face)	mm	32.00 ± 0.5
	iii) Length	mm	635 (approx)
6.	Slip strength	KN	64.1155 (95% of conductor UTS-67.49kN)
7.	Maximum resistance of the compressed unit expressed, as percentage of the resistance of equivalent length of bare conductor.	%	75
8.	Minimum corona Extinction voltage kV (rms) under dry condition	KV	320
9.	Maximum Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) under dry condition	Micro Volts	1000

6. Repair sleeve for AAAC PANTHER Conductor

S.No.	Description	Unit	Particulars/ Value
1.	Material		Aluminium of minimum purity 99.5% as per SS standards 4240813 & 4240814
2.	Dimension of Aluminium sleeve Before compression		
	i) Inside diameter	mm	23.00 ± 0.5
	ii) Outside diameter	mm	38.00 ± 1.0
	iii) Length	mm	241 ± 5.0
3.	Dimensions of Aluminium Sleeve after compression		
	i) Outside dimension (Corner to corner)	mm	37.00 ± 0.5
	ii) Outside dimension (face to face)	mm	32.00 ± 0.5
	iii) Length	mm	241 (Approx.)
4.	Minimum corona Extinction voltage kV (rms) under dry condition	KV	320
5.	Maximum Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) under dry condition	Micro Volts	1000

7. Vibration Damper for AAAC Panther conductor

S.No.	Description	Unit	Particulars/ Value
1.	Type of Damper		4R-Stockbridge type
2.	Materials of components		

	a) Damper masses		Cast iron/mild steel/Zinc alloy duly hot dip galvanised
	b) Clamp		Aluminum alloy 4600
	c) Messenger cable		High tensile strength galvanized steel
3.	Number of strands in stranded messenger cable	Nos.	19
4.	Minimum ultimate tensile strength of stranded messenger cable	Kg/mm2	135
5.	Slip strength of stranded messenger cable (mass pull off)	KN	5
6.	Slipping strength of damper clamp		
	(a) Before fatigue test	KN	2.5
	(b) After fatigue test	KN	2
7.	Resonance frequencies range	Hz	5 to 40
8.	Maximum magnetic power loss per vibration damper watts for 600 amps, 50 Hz Alternating Current	Watts	1
9.	Minimum corona Extinction voltage kV (rms) under dry condition	KV	320
10.	Maximum Radio Interference Voltage (RIV) at 1 MHz for phase to earth voltage of 305 kV (rms) under dry condition	Micro Volts	1000
11.	Percentage variation in reactance after fatigue test in comparison with that. before fatigue test	%	+/-40 (Maximum)
12.	Percentage variation in power dissipation after fatigue test in comparison with that before fatigue test	%	+/-40 (Maximum)
13.	Galvanising		
	a) Minimum weight of Zinc coating for steel parts	gm/m2	900
	b) Purity of Zinc used for galvanising	%	99.95 (IS 209) or 98.5(IS 13229)
	c) Min. No. of dips in standard preece test the ferrous parts can withstand.	Nos.	a) Fasteners: 4 dips of 1 minute b) Spring washers: 3 dips of 1 minute c) all others: 6 dips of 1 minute

ANNEXURE-IV

GUARANTEED TECHNICAL PARTICULARS/ TECHNICAL DATA SHEET

(TO BE FILLED BY BIDDER)

TYPICAL GUARANTEED TECHNICAL PARTICULARS OF SUSPENSION HARDWARE FITTINGS

Sl. No.	Description	Unit	Value guaranteed by the Bidder
1.	Name & address of Manufacturer		
2.	Address of Manufacturer		
3.	Drawing enclosed	Yes/No	
4.	Maximum magnetic power loss of suspension clamp at conductor / sub- conductor current of amperes (at steady state conductor temperature)	Watt	
5.	Slipping strength of suspension assembly (clamp torque Vs slip curve shall be enclosed)	kN	
6.	Particulars of standard/AGS Standard / AGS preformed armour rod set for suspension assembly		
	a) No. of rods per set	No.	
	b) Direction of lay		
	c) Overall length after fitting on conductor	mm	
	d) Actual length of each rod along its helix	mm	
	e) Diameter of each rod	mm	
	f) Tolerance in		
	i) Diameter of each rod	±mm	
	ii) Length of each rod	±mm	
	iii) Difference of length between the longest and shortest rod in a set	±mm	
	) Type of Aluminium alloy used for manufacture of PA rod set		
	h) UTS of each rod	Kg/mm2	
7.	Particulars of Elastomer (For AGS Clamp only)		
	a) Supplier of elastomer		
	b) Type of elastomer		
	c) Shore hardness of elastomer		
	Temperature range for which elastomer is designed		
	e) Moulded on insert		
8.	UTS of suspension clamp		
9.	Purity of Zinc used for galvanizing	%	
11.	Minimum corona extinction voltage under dry condition [for 400kV lines]	kV (rms)	
12.	Radio interference voltage at 1 Mhz for phase to earth voltage of 305 kV (dry condition) [for 400kV lines]	µV	
13.	Maximum permissible continuous operating temperature of		
	i) Clamp body	Deg. C	
	ii) Standard/AGS preformed rods	Deg. C	

GUARANTEED TECHNICAL PARTICULARS OF TENSION HARDWARE FITTINGS

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3.	Drawing enclosed	Yes/ No		
4.	Purity of aluminum used for aluminum sleeve	%		
5.	Material for steel sleeve			
	(i) Type of material with chemical composition			
	Range of Hardness of material (Brinnel Hardness)	BHN	From …….to ……..	
	(iii) Weight of zinc coating	gm/m2		
			Aluminium / Alloy	Steel
6.	Outside diameter of sleeve before compression	mm		
7.	Inside diameter of sleeve before compression	mm		
8.	Length of sleeve before compression			
9.	Dimensions of sleeve after compression			
	(a) Corner to Corner			
	(b) Surface to Surface			
10.	Length of sleeve after compression			
11.	Weight of sleeve			
	(a) Aluminium/ aluminum Alloy	kg		
	~~(b) Steel~~	~~kg~~		
	(c) Total	kg		
12.	Electrical resistance of dead end assembly as a percentage of equivalent length of Conductor	%		
13.	Slip strength of dead end assembly	kN		
14.	UTS of dead end assembly	kN		
15.	Purity of Zinc used for galvanizing	%		
16.	Design calculation of yoke plates and sag adjustment plate enclosed.	Yes / No		
17.	Minimum corona extinction voltage under dry condition [for 400kV lines]	kV (rms)		
18.	Radio interference voltage at 1 Mhz for phase to earth voltage of 305 kV (dry condition) [for 400kV lines]	µV		
19.	Maximum permissible continuous operating temperature of dead end assembly	Deg. C		

GUARANTEED TECHNICAL PARTICULARS OF MID SPAN COMPRESSION JOINT FOR HIGH PERFORMANCE (HP) CONDUCTORS-AAAC PANTHER

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3.	Drawing enclosed		Yes/No	
4.	Suitable for conductor size	mm		
5.	Purity of aluminium used for aluminium sleeve	%		
6.	Material for steel sleeve			
	(i) Type of material with chemical composition			
	(ii) Range of Hardness of material (Brinnel Hardness)	BHN	From …….to ……..	
	(iii) Weight of zinc coating	gm/m2		
			Aluminium / alloy	Steel
7.	Outside diameter of sleeve before compression	mm		
~~8.~~	~~Inside diameter of sleeve before compression~~	~~mm~~		
9.	Length of sleeve before compression			
10.	Dimensions of sleeve after compression			
	(a) Corner to Corner			
	(b) Surface to Surface			
11.	Length of sleeve after compression			
12.	Weight of sleeve			
	(a) Aluminium	kg		
	~~(b) Steel~~	~~kg~~		
	(c) Total	kg		
13.	Slip strength	kN		
14.	Resistance of the compressed unit expressed, as percentage of the resistivity of equivalent length of bare conductor.	%		
15.	Minimum Corona extinction voltage under dry condition [for 400kV lines]	kV (rms)		
16.	Radio interference voltage at 1 MHz for phase to earth voltage of 305 kV under dry condition[for 400kV lines]	µV		
17.	Maximum permissible continuous operating temperature of mid span compression joint	Deg. C		

GUARANTEED TECHNICAL PARTICULARS OF REPAIR SLEEVE FOR HIGH PERFORMANCE (HP) CONDUCTORS-AAAC PANTHER

Sl. No.	Description	Unit	Value guaranteed by the Bidder
1.	Name of Manufacturer		
2.	Address of Manufacturer		
3.	Drawing enclosed	Yes/No	
4.	Suitable for conductor size	mm	
5.	Purity of Aluminium / Al Alloy type	%	
6.	Dimension of sleeve before compression		
	i) Inside diameter of sleeve	mm	
	ii) Outside dimensions of sleeve	mm	
	iii) Length of sleeve	mm	
7.	Dimension of sleeve after compression		
	i) Corner to Corner	mm	
	ii) Surface to Surface	mm	
	iii) Length of sleeve	mm	
8.	Weight of sleeve	Kg	
9.	Minimum Corona extinction voltage under dry condition [for 400kV lines]	kV (rms)	
10.	Radio interference voltage at 1 MHz for phase to earth voltage of 305 kV dry condition) [for 400kV lines]	μV	
11.	Maximum permissible continuous operating temperature of Repair Sleeve	Deg. C	

NOTE: Tolerances, wherever applicable, shall also be specified.

GUARANTEED TECHNICAL PARTICULARS OF VIBRATION DAMPER FOR HIGH PERFORMANCE (HP) CONDUCTORS-AAAC PANTHER

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3	Drawing enclosed			
	(a) Design Drawing	YES/ NO		
	(b) Placement Chart	YES/ NO		
4.	Suitable for conductor size	mm		
5.	Total weight of one damper	kg		
			Right	Left
6.	Diameter of each damper mass	mm		
7.	Length of each damper mass	mm		
8.	Weight of each damper mass	kg		
9.	Material of damper masses			
10.	Material of clamp			
11.	Material of the stranded messenger cable			
12.	Number of strands in stranded messenger cable			
13.	Lay ratio of stranded messenger cable			
14.	Minimum ultimate tensile strength of stranded messenger cable	Kg/m m2		
15.	Slip strength of stranded messenger cable (mass pull off)	kN		
			Right	Left
16.	Resonance frequencies			
	(a) First frequency	Hz		
	(b) Second frequency	Hz		
17	Designed clamping torque	Kg-m		
18.	Slipping strength of damper clamp			
	(a) Before fatigue test	kN		
	(b) After fatigue test	kN		
19.	Magnetic power loss per vibration damper watts for ----Amps, 50 Hz Alternating Current [average continuous operating current]	watts		
20.	Minimum corona Extinction voltage kV (rms) under dry condition [for 400kV lines]	kV		
21.	Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) Microvolts under dry condition [for 400kV lines]	µV		
22.	Maximum permissible continuous operating temperature of Vibration Damper	Deg. C		
23.	Percentage variation in reactance after fatigue test in comparison with that . before fatigue test	%		
24.	Percentage variation in power dissipation after fatigue test in comparison with that before fatigue test	%		

NOTE: Tolerances, wherever applicable, shall also be specified.

GUARANTEED TECHNICAL PARTICULARS OF BUNDLE SPACER FOR HT/HTLS CONDUCTOR (IF APPLICABLE)

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3.	Drawing enclosed			
	(a) Design Drawing		YES / NO	
	(b) Placement Chart		YES / NO	
4	Suitable for conductor size	mm		
5.	Material / Manufacturing process of component parts			
			Material	Manufacturing Process
	(a) Insert			
	(b) Main body			
	(c) Retaining rods (if any)			
6.	Retaining rods (if used)			
	(a) Type of alloy used			
	(b) Number of retaining rods used for each spacer	no.		
	(c) Diameter	mm		
	(d) Length	mm		
	(e) Weight	kg		
7.	Elastomer			
	(a) Supplier			
	(b) Type			
	(c) Moulded on insert			
	(d) Shore hardness			
	(e) Thickness on insert	mm		
	(f) Temp. range for which designed	°C		
8.	Minimum ultimate tensile strength of spacer			
	(a) Compressive load	kN		
	(b) Tensile load	kN		
9.	Weight of Spacer	kg		
10.	Designed clamping torque(if applicable)	Kg m		
			Before Vibration	After Vibration
11.	Slipping strength of spacer clamp	kN		
12.	Magnetic power loss per spacer for ---- A, 50 Hz Alternating Current (at steady state conductor temperature)	Watt s		
			Maximum	Minimum
13.	Electrical resistance of elastomer cushioned spacer	oh m		
14.	Minimum corona Extinction voltage kV (rms) under dry condition [for 400kV lines]	kV		
15.	Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) Microvolts under dry condition [for 400kV lines]	µV		
16.	Maximum permissible continuous operating temperature of Bundle spacer	Deg. C		

NOTE: Tolerances, wherever applicable, shall also be specified.

GUARANTEED TECHNICAL PARTICULARS OF RIGID SPACER FOR JUMPER FOR HTLS CONDUCTOR

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3.	Drawing enclosed			
	(a) Design Drawing	YES / NO		
	(b) Placement Chart	YES / NO		
4	Suitable for conductor size	mm		
5.	Material of component parts			
	(a) Clamp			
	(b) Main body			
6.	Manufacturing process for			
	(a) Clamp			
	(b) Main body			
	(e) Weight	kg		
7.	Elastomer			
	(a) Supplier			
	(b) Type			
	(c) Moulded on insert			
	(d) Shore hardness			
	(e) Thickness on insert	mm		
	(f) Temp. range for which designed	°C		
8.	Minimum ultimate tensile strength of spacer			
	(a) Compressive load	kN		
	(b) Tensile load	kN		
9.	Weight of Spacer	kg		
10.	Designed clamping torque(if applicable)	kg.m		
11.	Slipping strength of spacer clamp	kN		
12.	Magnetic power loss per spacer for Watts ---Amps, 50 Hz Alternating Current (at steady state conductor temperature)	watt		
			Maximum	Minimum
12.	Electrical resistance of elastomer cushioned spacer	ohm		……
13.	Minimum corona Extinction voltage kV (rms) under dry condition [for 400kV lines]	kV (rms)		
14.	Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) Microvolts under dry condition [for 400kV lines]	µV		
15.	Maximum permissible continuous operating temperature of rigid spacer	Deg. C		

NOTE: Tolerances, wherever applicable, shall also be specified.

GUARANTEED TECHNICAL PARTICULARS OF SPACER DAMPER FOR HTLS CONDUCTOR (IF APPLICABLE)

Sl. No.	Description	Unit	Value guaranteed by the Bidder	
1.	Name of Manufacturer			
2.	Address of Manufacturer			
3.	Drawing enclosed			
	(a) Design Drawing	YES/ NO		
	(b) Placement Chart	YES/ NO		
4	Suitable for conductor size	mm		
5.	Material of component parts			
	(a) Clamp			
	(b) Main body			
6.	Type of Clamps			
7.	Type of Damping element			
8.	Manufacturing process for			
	(a) Clamp			
	(b) Main body			
	(e) Weight	kg		
9.	Elastomer			
	(a) Supplier			
	(b) Type			
	(c) Moulded on insert			
	(d) Shore hardness			
	(e) Thickness on insert	mm		
	(f) Temp. range for which designed	°C		
10.	Minimum ultimate tensile strength of spacer			
	(a) Compressive load	kN		
	(b) Tensile load	kN		
11.	Weight of Spacer	kg		
12.	Designed clamping torque(if applicable)	kg. m		
13.	Slipping strength of spacer clamp	kN		
14.	Magnetic power loss per spacer for Watts 1574 Amps, 50 Hz Alternating Current	watt		
			Maximum	Minimum
15.	Electrical resistance of elastomer cushioned spacer	ohm		
16.	Minimum corona Extinction voltage kV (rms) under dry condition [for 400kV lines]	kV (rms)		
17.	Radio Interference Voltage at 1 MHz for phase to earth voltage of 305 kV (rms) Microvolts under dry condition [for 400kV lines]	µV		
18.	Maximum permissible continuous operating temperature of spacer damper	Deg. C		

NOTE: Tolerances, wherever applicable, shall also be specified.